



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (Biotechnology) (CBCS) Degree Examinations

December 2023

BT2102-1 – BIOPROCESS PRINCIPLES & CALCULATIONS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following is not a correct method of writing unit for time?
A) min
B) minutes
C) minute
D) mins
- A liquid mixture has 2 components: A & B. Their composition is 25% and 75% by mole respectively. Here, A & B are isomers having the same molecular weight. Mass percent of A is:
A) 25
B) 75
C) 33.33
D) Data insufficient
- One of the following statements is wrong. Identify it.
S1: Density of water at NTP is 1 kg/L
S2: Viscosity of water at NTP is 1 cP
S3: Density of air at NTP is 784 kg/m³
S4: At NTP, Temp. = 20°C & Pressure = 1 atm
A) S1
B) S2
C) S3
D) S4
- You are given a 2M NaOH solution. NaOH shows a valency of one. What is the concentration of NaOH solution in terms of Normality?
A) 1 N
B) 2 N
C) 1/40 N
D) 2/40 N
- "If two thermodynamic systems are each in thermal equilibrium with a third one, then they are in thermal equilibrium with each other." Which law is it?
A) Zeroth Law of Thermodynamics
B) First Law of Thermodynamics
C) Second Law of Thermodynamics
D) Third Law of Thermodynamics
- In which of these processes, change in Internal Energy is zero for a process involving ideal gas?
A) Isobaric process
B) Isothermal process
C) Isochoric process
D) Adiabatic process
- Which of the following is not a state function?
A) Work
B) Internal Energy
C) Enthalpy
D) Entropy
- If the Specific heat (Cp) of an ideal gas is 28.316, then what is the adiabatic constant γ ? Note that the adiabatic constant is defined as ratio of specific heats Cp and Cv.
A) 8.316
B) 20
C) 1.416
D) 1
- The useful product stream in an extraction operation is termed as:
A) Feed stream
B) Solvent stream
C) Extract
D) Raffinate
- Drying operation can be termed as:
A) Solid-Liquid operation
B) Liquid-Liquid operation
C) Gas-Liquid operation
D) Vapor-Liquid-Solid operation
- While solving the material balance equations for Crystallization process, how do you estimate the concentration of hydrated salt in crystal component of the product stream?
A) Using water of hydration data
B) Using solubility data
C) Using humidity chart
D) Using specific heat data

12. Which of these represent as a recycle operation?
 A) Reflux in distillation B) Evaporation of water in crystallization
 C) A part of feed in drying is sent to product D) Immiscibility of solvents in extraction
13. Which of these is best example for energy transfer process without reaction?
 A) ATP hydrolysis B) Any exothermic reaction
 C) Cooling of fermentation fluid D) Incineration of biomass
14. Energy required to completely evaporate one kg of liquid water at its boiling point:
 A) Zero B) 4.2 kJ
 C) 333 kJ D) 2442.5 kJ
15. Equation for specific heat, $C_p = a + bT + cT^2 + dT^3$ is usually followed by:
 A) Gases and few Liquids only B) All types of Liquids
 C) Solids only D) All substances
16. While performing energy balance in fermenters under steady state operation, which law of thermodynamics is applied?
 A) Zeroth Law B) First Law
 C) Second Law D) Third Law
17. Reaction: $\text{CO} + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}_2$. One mole of CO fed with 2 moles of O_2 . Identify correct statement.
 A) CO is limiting reactant B) O_2 is limiting reactant
 C) CO is excess reactant D) 2 moles of CO_2 are produced.
18. Degrees of reduction for Glucose is
 A) Zero B) 4
 C) 24 D) 6
19. The yield termed as $Y_{x/s}$ represents
 A) Biomass yield per mole of carbon B) Product yield per mole of oxygen
 C) Biomass yield per mole of oxygen D) Product yield per mole of carbon
20. Orsat analysis is useful for
 A) Fuel gas analysis B) Biomass composition analysis
 C) Excess air supply D) Product or Flue gas composition analysis

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) A natural gas has the following composition by volume: $\text{CH}_4 - 82\%$, $\text{C}_2\text{H}_6 - 12\%$ and $\text{N}_2 - 6\%$. Calculate the density of gas in kg/m^3 at 288 K and 1 atm pressure.	08	L*3	1	2,3
	b) Carnot's cycle represents functioning of an ideal Carnot engine. With sketch explain the Carnot's cycle and also state the Carnot's two principles on efficiencies of reversible and irreversible engine.	08	L2	2	2
2.	a) One kmol of CO_2 occupies a volume 0.381 m^3 at 313 K. Compare the pressure of CO_2 using ideal gas equation and van der Waal's equation. Given: $a = 0.365 \text{ SI unit}$, $b = 4.28 \times 10^{-5} \text{ SI Unit}$.	08	L3	1	1,2
	b) For an open system with a flow process, change in enthalpy is given as the difference in heat supplied to workdone. Prove this statement by applying 1 st law of thermodynamics to a flow process when change in kinetic energy and change in potential energy are negligible.	08	L3	2	1,2
3.	a) A mixture of water and ethanol has composition 50% by mole each. Convert this composition to mass%. Also determine how much quantity (in kg) of ethanol is present in 10 kg of solution.	08	L3	1	1,2
	b) An ideal gas is confined within a closed cylinder and the system may be considered to be under adiabatic condition. If adiabatic constant for the ideal gas is 1.4, then prove that $P_1 V_1^{1.4} = P_2 V_2^{1.4}$ given that suffixes 1 and 2 refer initial and final conditions respectively.	08	L3	2	1,2

Unit – II

4. a) A mixed acid (400 kg) of composition 40% H_2SO_4 , 45% HNO_3 and rest H_2O is to be produced by strengthening waste acid of composition 30% H_2SO_4 , 36% HNO_3 and 34% H_2O by weight. Concentrated sulfuric acid of strength 95% and concentrated nitric acid containing 80% are available for this purpose. Write the mass balance equations and find how many kg of spent acid and concentrated acids are to be mixed together?

08 L3 3 2,3

- b) Steam is used to heat nutrient medium in a continuous-flow process. Saturated steam at 150°C enters a coil on the outside of the heating vessel and is completely condensed. Medium enters the vessel at 15°C and leaves at 44°C . Heat losses from the jacket to the surroundings are estimated as 0.22 kW. If the flow rate of medium is $3250 \text{ kg}\cdot\text{h}^{-1}$, estimate the quantity of steam required?

Data: Heat capacity of medium $C_p = 3.7 \text{ kJ}\cdot\text{kg}^{-1}\cdot^\circ\text{C}^{-1}$,Enthalpy of steam (150°C) = 2113.1 kJ/kg .

08 L4 4 2,3

5. a) A saturated solution of MgSO_4 at 80°C is cooled to 30°C in a crystallizer. During cooling 10% water is lost by evaporation. Calculate the quantity of the original solution to be fed to obtain 500 kg of MgSO_4 . Solubility of salt in water (g MgSO_4 per 100 g H_2O) are: 64 at 80°C and 40 at 30°C .

08 L4 3 2,3

- b) A distillation column separates 100 mol/h of a mixture containing 45 mol% benzene and rest toluene. The product D recovered from the condenser at the top of the column contains 90 mol% benzene, and the bottom W from the column contains 95 mol% toluene. The vapor V entering the condenser from the top of the column is 75 mol/h. A portion of the product from the condenser is returned to the column as reflux R, and the rest is withdrawn as the final product D. Assume that V, R, and D are identical in composition since V is condensed completely. Find the ratio of the amount refluxed R to the product withdrawn D.

08 L3 4 2,3

6. a) A 100 kg feed consisting of 27.8% acetone (A), 72.2% chloroform (B), by weight is to be batch extracted with a mixed solvent at 25°C . This mixed solvent of unknown composition is known to contain water (C) and acetic acid (D). The mixture of the feed and the solvent is shaken well, allowed to attain equilibrium and separated into two layers. The composition of two layers is given below.

Composition (%)		A	B	C	D
Layer	Upper layer	7.5	3.5	57.4	31.6
	Lower layer	20.3	67.3	2.8	9.6

Find

- the quantities of two layers
- solvent to feed weight ratio
- composition of solvent.

08 L3 3 2,3

- b) A natural gas has the following composition on mole basis:

 $\text{CH}_4 = 84\%$ $\text{C}_2\text{H}_6 = 13\%$ $\text{N}_2 = 3\%$

Calculate the heat to be added to heat one kmol of natural gas from 298 K to 523 K using the heat capacity data given below:

 $C_p^\circ = a + bT$, $\text{kJ}/(\text{kmol}\cdot\text{K})$

Gas	a	$b \times 10^3$
CH_4	19.2494	52.1135
C_2H_6	5.4129	178.0872
N_2	29.5909	- 5.141

08 L4 4 2,3

Unit – III

7. a) Hydrolysis reaction of p-Nitrophenol phosphate (PNPP, Mol. Wt = 219) can be accomplished by the enzyme acid phosphatase to produce products p-Nitrophenol (PNP, Mol. wt. = 139) and phosphate (P, Mol. wt. = 98) as per the following reaction scheme:



In a batch reactor, with sufficient amount of enzyme, an aqueous solution of PNPP at a concentration of 10mM produces 973 mg/L of PNP.

- Identify limiting reactant, excess reactant
- Calculate fractional molar conversion, yield (weight basis).
- Determine concentration of phosphate (P) in g/L.

08 L4 5 2,3

- b) Biomass is to be used as fuel for combustion. If you were provided with samples of biomass, then how would you analyze and report its physical composition by making use of Proximate and Ultimate analysis? Make use of both the methods and demonstrate the procedures.

08 L2 5 2

8. a) Production of single-cell protein from hexadecane is described by the following reaction equation:



Where $\text{CH}_{1.66}\text{O}_{0.27}\text{N}_{0.20}$ represents the biomass. If RQ= 0.43, determine the stoichiometric coefficients.

08 L3 5 1,2,3

- b) The alcohol dehydrogenase reaction removes ethanol from the blood.



Heat of formation data ΔH_f (kJ/mol): NAD (-10.3); NADH (-41.4); Ethanol (-290.8); Acetaldehyde (-214.0).

- Calculate standard heat of reaction.
- If 10.0 g of ethanol is completely converted to acetaldehyde by this reaction, how much heat is produced or consumed?

08 L4 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
